Research Project periodic log entry form

Project ID : 39
Student ID : IT22182678
Name : Rashani K.G.M.
Individual Component : ML-Based Quality Predition and Requirements Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | Work |
|---|---|
| 01.04.2026 | Started planning the basic workflow and component structure of the system. |
| 03.04.2026 | Defined how the ML-based test quality predition system. |
| 05.04.2026 | Designed sample predition senarios such as single test case quality scoring, batch predictions, RTM generation |
| 07.04.2026 | Worked on structuring the backend communication flow between FastAPI and React frontend. |
| 11.04.2026 | Got team meeting with supervisor to disscuss progress. |

Supervisor's comments: Currently working on the Component

Student processing well in research project : (Yes) / No

If no:

Project work to be correct out before next evaluation period.

code based checked : (Yes) / No

RICHARD

Signature:

Date: 20/04/2026

Research Project periodic log entry form.

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML - based Quality Prediction and Requirements Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | Work |
|---|---|
| 20.04.2026 | working on designing the data input structure for the ML based key features and parameters. |
| 21.04.2026 | Integrated the ML model with the backend and tested basic prediction functionality. |
| 23.04.2026 | worked on connecting the frontend and displaying prediction results |
| 27.04.2026 | Reviewed system functionality and fixed minor bugs related to API responses |
| 29.04.2026 | improved the accuray of the ML model by tuning parameters and testing with additional dataset. |

Supervisor's Comments - Currently working on the 50% Component.

Student processing well in research project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

Combine the backend with the front UI's and show the demo.

code base checked : (Yes) / No

signature :

Date : 30/04/2026

Research Project periodic log entry form

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML - based Quality Prediction and Requirements Traceability Matrix (RTM)

Summary of RP Work carried out during the reporting period

| Date | Work |
|---|---|
| 04.05.2026 | Designed the Test Quality Prediction interface and finalized the layout for file upload and manual input functionalities. |
| 08.05.2026 | Implemented the test case upload feature to accept JSON, CSV and Excel files for quality analysis. |
| 11.05.2026 | Developed the ML prediction workflow and integrated it with the backend services. |
| 13.05.2026 | Configured quality score calculations based on assertion strength, code coverage, boundary testing and error handling metrics. |
| 15.05.2026 | Designed and implemented the Quality prediction Results dashboard with graphical score visualization. |

Supervisor's comments -

Student processing well in research project : (Yes) / No

If no:

Project work to be correct out before next evaluation period :

code based checked : (Yes) / No

Signature :

Date : 30/05/2026

RICHARD

Research project periodic log entry form.

project ID : 39

student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML-based Quality prediction and Requirements Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period

| Date | Work |
|---|---|
| 18.05.2026 | Implemented the Requirements Traceability Matrix module to map requirements with corresponding test cases. |
| 21.05.2026 | Implemented Similarity score calculation and coverage status identification for requirement test case mapping |
| 25.05.2026 | Added RTM analytics including total links, covered requirements, partial coverage |
| 27.05.2026 | Implemented PDF report generation functionality for RTM analysis and prediction results. |
| 29.05.2026 | Got team meeting with supervisor to discuss progress. |

Supervisor's comments -

Student processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code base checked : (Yes) / No

Signature :

Date : 30/05/2026

RICHARD

Research project periodic log entry form.

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality prediction and Requirement traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | Work. |
|---|---|
| 01.06.2026 | Reviewed component 1 - Intelligent user story Generator service and its available API endpoints |
| 02.06.2026 | Integrated the existing component 1 API with the RTM Matrix page without modifying the component 1 source code. |
| 03.06.2026 | Retrivied Requirement and user story information such as Requirement ID, Requirement text, Requirement Type, Requirement status, user story ID, Priority and Acceptance criteria. |
| 04.06.2026 | Reviewed component 2 - Intelligent Test case Generation and identified the available test case and traceability endpoints. |
| 05.06.2026 | Integrated component 2 test case data with the RTM Matrix page worked on estabilishing the relationship between user story ID and coresponding Test cases. |

RICHARD

supervisor's comments -

Student processing well in research project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

Code base checked : (Yes) / No

Signature :

Date : 30/05/2026

RICHARD

Research project periodic log entry form.

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Tracebility Matrix (RTM)

Summary of RP work carried out during the reporting.

| Date | Work |
|---|---|
| 08.06.2026 | Updated the RTM matrix structure to display requirement, user story and Test case in single traceability view. |
| 10.06.2026 | Implemented the ~~tr~~ retrieval and display of requirement and story details from component 1. |
| 11.06.2026 | Verified that API response was correctly mapped to the corresponding RTM fields. Implemented the retrieval of test case information from Component 2. |
| 12.06.2026 | Filtered the API response to obtain the required Test case ID, Test Description and Test Pass / Fail status. |
| 13.06.2026 | Worked on mapping user story IDs with their relevant test cases. Ensured that each user story could be associated with the appropriate test cases in the RTM Matrix. |

RICHARD

Supervisor's comment -

Student processing well in research project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

code based checked : (Yes) / No

Signature : ~~[illegible]~~

Date : 29/06/2026

RICHARD

Date...../...../.....  No:.....

Research Project periodic log entry form

project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Tracebility Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | work |
|---|---|
| 15.05.2026 | Tested the comple RTM data flow from component 1 to component 2 to RTM Matrix. Fixed data mapping and rendering issues. |
| 16.05.2026 | Identified important quality characteristics such as completeness, requirement coverage, specify, clarity, expected results. and ambiguity. |
| 17.05.2026 | Designed the Quality Scoring approach for test cases. Defined a weighted quality score using test case characteristics. |
| 18.05.2026 | Given the quality prediction result for High, Medium and Low quality categories. |
| 19.06.2026 | Created a structured dataset for training the Quality prediction model. Designed features based on test case characteristics and incorporated Given, When, Then scenarios from test descriptions |

RICHARD

Date...../...../.....  No:.....

Supervisor's comment -

Student processing well in Research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code base checked : (Yes) / No

Signature :

Date : 29/06/2026

RICHARD

Research Project periodic log entry form.

project ID : 39

Student ID : IT22162678

Name : Rashani K.G.M.

Individual Component : ML based Quality prediction and Requirement Tracebility Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | work. |
|---|---|
| 22.06.2026 | Prepocessed the dataset and prepared the features required for maching learning. Generated appropiate quality labels based on the defined quality scoring. |
| 23.06.2026 | Implemented and trained a Random Forest classifier for test case quality prediction. Evaluated the model and reviewed the prediction probalities. |
| 25.06.2026 | Updated the Quality prediction page to use Given / when / Then test description from the RTM Matrix instead of using requirement text. |
| 26.06.2026 | Implemented the 'select a Test case' dropdown on the Quality Prediction page. Connected the dropdown to the available test cases. |
| 30.06.2026. | Integrated the trained ML model with the Quality prediction page. Implemented the prediction flow from Test case → Feature Extraction → ML model → Quality score → Quality level |

RICHARD

Supervisor's comment -

Student processing well in research project : (Yes) / NO

If no :

project work to be correct out before next evaluation period :

code based checked : (Yes) / NO

Signature : [redacted]

Date : 30/06/2026

RICHARD

Research project periodic log entry form:

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual component : ML based Quality prediction and Requirement Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | Work |
|---|---|
| 01.07.2026 | Enhanced the trained ML model with Quality prediction page. Implemented the prediction flow from Quality level |
| 02.07.2026 | Enhanced the Quality prediction results by displaying relevant information such as Quality score, Quality level, probability / confidence and important factors. |
| 03.07.2026 | Updated the Test Inventory page to display the quality prediction results. Modified the table to show relevant test case information and the predict results |
| 06.07.2026 | Tested the integration between the RTM Matrix, Quality prediction and Test Inventory pages. verified that test case data was consistently transfered between relevant pages. |
| 08.07.2026 | Fixed Issues related to API data retrieval, test case mapping, drop down population and quality prediction results. |

RICHARD

Supervisor's comment -

Student processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code based checked : (Yes) / No

Signature :

Date : 30/07/2026

RICHARD

Date: / /  No:

Research project periodic log entry form.

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality prediction and Requirement Tracebility Matrix (RTM)

Summary of RP work carried out during the reporting period

| Date | Work |
|---|---|
| 09.07.2026 | worked on the Github code and Branch coverage functionality. Investigated project detection issues and improved support for different project types |
| 10.07.2026 | Tested the application using different repositories and project structures. Investigated coverage generation errors and improved the handling of project detection, testing and coverage report generation. |
| 13.07.2026 | performed final testing of the complete components verified the RTM Matrix, Quality prediction, Test Inventory, API integrations and code coverage function |
| 14.07.2026 | Reviewed the existing coverage Gaps functionality and studied how requirements, acceptance criteria, test cases and code coverage. |
| 15.07.2026 | Implement the logic for identifying requirements with partial or missing test coverage. |

RICHARD

Date: / /  No:

Supervisor's comment -

Student processing well in research project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

code based checked : (Yes) / No

Signature :

Date : 30/07/2026

RICHARD

Research project periodic log entry form.

project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality prediction and Requirement Tracebility Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | work |
|---|---|
| 16.07.2026 | Integrated requirement and test coverage information into the coverage Gaps page. Displayed relevant requirement details, coverage status and risk level. |
| 17.07.2026 | worked on Risk based prioritization for coverage gaps. Considered factors ~~seh~~ such as business priority, acceptance criteria coverage. |
| 20.07.2026 | Tested the coverage Gaps page and verified that requirements were correctly categorized as critical, Medium or Low. based on their coverage |
| 22.07.2026 | Fixed data mapping and and UI issues identified during testing. |
| 24.07.2026 | Desined the Intelligent Test improvement and Recommondation feature to identify weaknesses in test cases based on their Quality prediction result. |

RICHARD

Supervisor's comment -

Student processing well in research project : (Yes) / NO

IF no :

project work to be correct out before next evaluation period :

code based checked : (Yes) / NO

Signature :

Date : 30/07/2026

RICHARD

Research project periodic log entry form

project ID : 39

Student ID : IT22162678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | Work |
|---|---|
| 27.07.2026 | Implemented the analysis of test case quality factors such as completeness, specificity and ambiguity |
| 29.07.2026 | Developed the Resolve Coverage Gap functionality. Desined the workflow to identify uncovered acceptance criteria and recommend appropiate. |
| 31.07.2026 | Implemented the generation of improved Given/when/Then test cases based on the selected requirement |
| 03.08.2026 | Integrated the improved test cases with the existing Test inventory and RTM Matrix functionality. |
| 05.08.2026 | Verified that generated test cases maintained the correct relationship between requirements, user stories and test cases. |

RICHARD

Supervisor's comment -

Student processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code base checked : (Yes) / No

Signature :

Date : 28/08/2026

RICHARD

Research project periodic log entry form

Project ID : 39

Student ID : IT22182678

Name : Roshani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Traceability Matrix (RTM)

Summary of RP work carried out during the reporting period.

| Date | work |
|---|---|
| 06.08.2026 | Designed a Test Quality Analytics and Insights section to provide an overall view of test case quality and testing effectiveness. |
| 10.08.2026 | Implemented quality-related metrics such as total test cases, average quality score, high/medium/low quality distribution |
| 12.08.2026 | Developed visual representations for quality and coverage information using suitable charts and dashboard components. |
| 13.08.2026 | Integrated quality prediction, test coverage and RTM information to provide a consolidated view of the overall testing quality. |
| 14.08.2026 | Reviewed the analytics functionality and refined the UI/UX to make the information easier to understand. |

RICHARD

supervisor's comment -

Student processing well in research project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

code base checked : (Yes) / No

Signature :

Date : 28/08/2026

RICHARD

Research project periodic log entry form

project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Tracebility Matrix (RTM)

Summary RP work and carried out during the reporting period

| Date | Work |
|---|---|
| 17.08.2026 | Performed ent-to-end integration testing of the complete component, including RTM Matrix, Test Inventory, Quality Prediction. |
| 18.08.2026 | Tested the communication between the frontend and backend services and verified the API responses used by the different pages. |
| 20.08.2026 | Tested the ML based quality prediction workflow using different Given/When/Then test cases verified the generated quality scores, quality leves. |
| 24.08.2026 | Performed functional testing of the coverage Gap and Intelligent Test improvement features. |
| 25.08.2026 | Conducted final system testing and fixed remaining UI, API, data mapping and integration issues. Reviewed the complete component to ensure that all implemented features worked together correctly and were ready for the final research demonstration |

RICHARD

Supervisor's comment :

Student processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code base checked : (Yes) / No

Signature :

Date : 28/08/2026

RICHARD

Research project periodic log entry form

project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality prediction and Requirment Tracebility Matrix (RTM)

Summary RP work and carried out during the reporting period.

| Date | work |
|---|---|
| 01. 09. 2026 | Reviewed the backend structure and identified the main components that required testing. Examined the API endpoints and the integration between backend services. |
| 03. 09. 2026 | Tested the backend API endpoints to verify that they returned the expected responses. checked requested parameters, response formats and error messages. |
| 07. 09. 2026 | Tested the input validation mechanisms using valid, invalid, and incomplete data. Verified how the backend handled missing fields. |
| 09. 09. 2026 | Evaluated the backend error-handling mechanisms by testing different failure scenarios. |
| 11. 09. 2026 | Documented the results of the backend testing activities and identified issues that required further investigation. |

Supervisor's Comment :

Student Processing well in research Project : (Yes) / No

If no :

Project work to be correct out before next evaluation period :

code base checked : Yes / No

Signature :

Date :

Research project periodic log entry form.

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Tracebility Matrix (RTM)

Summary RP work and carried out during the reporting period.

| Date | work. |
|---|---|
| 14.09.2026 | Reviewed the machine learning based test quality prediction component and its role in evaluating the quality of test cases. |
| 17.09.2026 | Examined the test case data used by the quality prediction component. checked the data structure and input values to identify missing information. |
| 21.09.2026 | Tested the quality prediction workflow using different test case inputs. Reviewed the generated quality scores and predicted quality categories. |
| 24.09.2026 | Analysed the prediction results to identify potential inconsistencies and unexpected outputs. |
| 28.09.2026 | Recorded the outcomes of the quality prediction tests and documented the issues identified during validations. |

RICHARD

Supervisor's comment :

Student Processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

code base checked : Yes / No

Signature :

Date :

RICHARD

Research project periodic log entry form

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Tracebility Matrix (RTM)

Summary RP work and carried out during the reporting period

| Date | work |
|---|---|
| 02.10.2026 | Reviewed the requirement-to-test-case mapping process using available project data. Checked whether the generated traceability information represented the expected relationships. |
| 04.10.2026 | Tested the requirement to test case mapping process using available project data. checked whether the generated traceability informations |
| 06.10.2026 | Examined the generated RTM output to verify the structure and consistency of the displayed Traceability information. |
| 08.10.2026 | Tested scenarios involving missing or incomplete requirement and test case information. Reviewed how the system handled these scenarios. and |
| 09.10.2026 | Documented the RTM testing results and reviewed the identified mapping issues. |

RICHARD

Supervisor's comment :

Student Processing well in research project : (Yes) / No

If no :

project work to be correct out before next evaluation period :

Code base checked : Yes / NO

Signature:

Date :

RICHARD

Research project periodic log entry form

Project ID : 39

Student ID : IT22182678

Name : Rashani K.G.M.

Individual Component : ML based Quality Prediction and Requirement Traceblity Matrix (RTM)

Summary RP work and Carried out during the reporting period.

| Date | Work |
|---|---|
| 12.10.2026 | Tested the integration between the frontend and backend components. Verified that requests were processed correctly. |
| 13.10.2026 | Tested the main system workflow, including test quality Prediction and RTM generation. Reviewed the interaction between the relevant compones to identify integration issues. |
| 14.10.2026 | Retested the issues identified during Previous testing activities after the relevant corrections were made. |
| 15.10.2026 | Organized the testing observations, validation results and identified issues. Reviewed the technical documentation to ensure that the testing activities |
| 16.10.2026 | Conducted a final review of the research implementation and testing activities completed during final phase |

RICHARD

Supervisor's comment :

Student processing well in research project : (Yes) / NO

If no :

Project work to be correct out before next evaluation period :

code base checked : Yes / NO

Signature :

Date :

RICHARD